CLASS 10 · CBSE · NCERT MATHEMATICS · SHORT NOTES

Chapter 13: Statistics

Topic-wise short notes with solved examples

This chapter finds the three measures of central tendency, the mean, the mode and the median, for data that has been grouped into class intervals.

1 Grouped data: the terms

- **Class interval**: a group such as 10–20. Here 10 is the **lower limit** and 20 the **upper limit**.
- **Class size** h = upper limit − lower limit.
- **Class mark** x_i = $\dfrac{\text{lower limit} + \text{upper limit}}{2}$. It stands for all the values in the class.
- **Frequency** f_i: the number of observations in the class. $\sum f_i = n$ is the total number of observations.
- **Cumulative frequency** (cf): the running total of the frequencies.
- The classes must be **continuous** (10–20, 20–30, ...) before the mode and median formulas are used. If they are 10–19, 20–29, ..., subtract 0.5 from each lower limit and add 0.5 to each upper limit.

2 Mean of grouped data

- **Direct method**

$$\bar{x} = \frac{\sum f_i x_i}{\sum f_i}$$

- **Assumed mean method**: choose a class mark a near the middle, and let $d_i = x_i - a$.

$$\bar{x} = a + \frac{\sum f_i d_i}{\sum f_i}$$

- **Step-deviation method**: let $u_i = \dfrac{x_i - a}{h}$.

$$\bar{x} = a + \left(\frac{\sum f_i u_i}{\sum f_i}\right) \times h$$

- All three methods give the same mean. Use the direct method for small numbers and the step-deviation method when the class marks are large and the class sizes are equal.

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Example 2.1 Find the mean of the following data by all three methods.

Class	0–10	10–20	20–30	30–40	40–50
Frequency	5	8	15	16	6

SOLUTION

Take $a = 25$ and $h = 10$.

Class	f_i	x_i	$f_i x_i$	$d_i = x_i - 25$	$f_i d_i$	u_i	$f_i u_i$
0–10	5	5	25	-20	-100	-2	-10
10–20	8	15	120	-10	-80	-1	-8
20–30	15	25	375	0	0	0	0
30–40	16	35	560	10	160	1	16
40–50	6	45	270	20	120	2	12
Total	50		1350		100		10

Direct method: $\bar{x} = \dfrac{1350}{50} = 27$

Assumed mean method: $\bar{x} = 25 + \dfrac{100}{50} = 27$

Step-deviation method: $\bar{x} = 25 + \dfrac{10}{50} \times 10 = 27$

Mean $= 27$

Example 2.2 The mean of the following data is 27. Find the missing frequency p.

Class	0–10	10–20	20–30	30–40	40–50
Frequency	3	5	p	7	5

SOLUTION

$\sum f_i = 20 + p$ and $\sum f_i x_i = 15 + 75 + 25p + 245 + 225 = 560 + 25p$

$\dfrac{560 + 25p}{20 + p} = 27$, so $560 + 25p = 540 + 27p$

$p = 10$

3 Mode of grouped data

- The **modal class** is the class with the highest frequency.

$$\text{Mode} = l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right) \times h$$

- l = lower limit of the modal class, h = class size, f_1 = frequency of the modal class, f_0 = frequency of the class **before** it, f_2 = frequency of the class **after** it.

- The mode is the value that occurs most often. It lies inside the modal class.

Example 3.1 Find the mode of the data in the first example of topic 2.

SOLUTION

The highest frequency is 16, so the modal class is 30–40.

$l = 30,\ h = 10,\ f_1 = 16,\ f_0 = 15,\ f_2 = 6$

$$\text{Mode} = 30 + \frac{16 - 15}{32 - 15 - 6} \times 10 = 30 + \frac{10}{11}$$

Mode ≈ 30.91

4 Median of grouped data

- Write the cumulative frequencies and find $\frac{n}{2}$.
- The **median class** is the class whose cumulative frequency is just greater than (or equal to) $\frac{n}{2}$.

$$\text{Median} = l + \left(\frac{\frac{n}{2} - cf}{f}\right) \times h$$

- l = lower limit of the median class, n = total frequency, cf = cumulative frequency of the class **before** the median class, f = frequency of the median class, h = class size.
- The median divides the data into two equal halves.

Example 4.1 Find the median of the data in the first example of topic 2.

Class	0–10	10–20	20–30	30–40	40–50
Frequency	5	8	15	16	6
Cumulative frequency	5	13	28	44	50

SOLUTION

$n = 50$, so $\frac{n}{2} = 25$. The cumulative frequency just greater than 25 is 28, so the median class is 20–30.

$l = 20,\ cf = 13,\ f = 15,\ h = 10$

$$\text{Median} = 20 + \frac{25 - 13}{15} \times 10 = 20 + 8$$

Median $= 28$

Example 4.2 The median of the following data is 27.5 and the total frequency is 50. Find x and y.

Class	0–10	10–20	20–30	30–40	40–50	50–60
Frequency	4	x	16	12	y	4

SOLUTION

Total: $36 + x + y = 50$, so $x + y = 14$.

The median 27.5 lies in 20–30, so $l = 20$, $f = 16$, $cf = 4 + x$, $h = 10$, $\frac{n}{2} = 25$.

$27.5 = 20 + \frac{25 - (4 + x)}{16} \times 10$, so $7.5 \times 16 = 10(21 - x)$ and $21 - x = 12$

$x = 9$ **and** $y = 5$

5 Which measure to use

- **Mean**: uses every observation; best when the data has no extreme values. Example: average marks of a class.
- **Median**: the middle value; best when there are extreme values. Example: typical salary in a company.
- **Mode**: the most frequent value; best for finding what is most common. Example: the shoe size sold most.
- **Empirical relationship** between the three:

$$3 \times \text{Median} = \text{Mode} + 2 \times \text{Mean}$$

REMEMBER

The empirical relationship is approximate. For the data above, $3 \times 28 = 84$ and $30.91 + 2 \times 27 = 84.91$.

6 Common mistakes

- Using class marks in the median or mode formula. Both use the **lower limit** l of the class.
- Taking cf of the median class itself. Use the cumulative frequency of the class **before** it.
- Picking the median class by the highest frequency. It is chosen by $\frac{n}{2}$.
- Forgetting to multiply by h in the step-deviation method.
- Applying the formulas to classes that are not continuous.